



Assignment 01

Submitted By	Ijlal Ahmad
Submitted To	Ms. Iqra Naseem
Roll No	FL-21565
Date	14-05-2025
Subject	AI

Q1. Define Artificial Intelligence. Discuss its main goals and significance in today's world.

Definition:

Artificial Intelligence (AI) is the branch of computer science that aims to create machines or software that can perform tasks typically requiring human intelligence. These tasks include learning, problem-solving, decision-making, speech recognition, language understanding, and visual perception.

Main Goals of AI:

1. **Automation of Human Tasks:** Enabling machines to perform repetitive or complex tasks with little or no human intervention.
2. **Learning from Data:** Allowing systems to learn from experiences and improve over time.
3. **Reasoning and Problem Solving:** Developing systems that can draw logical conclusions or solve puzzles.
4. **Understanding Natural Language:** Enabling machines to understand, interpret, and respond in human language.
5. **Perception:** Enabling systems to perceive the world using inputs like cameras or sensors (e.g., facial recognition, object detection).
6. **Autonomy:** Creating systems that can operate independently, like self-driving cars or robotic assistants.

Significance in Today's World:

- **Healthcare:** AI helps in diagnosis (e.g., cancer detection), drug development, and personalized medicine.
- **Business & Finance:** Used for fraud detection, customer service (chatbots), and investment decisions.
- **Transportation:** Powers autonomous vehicles and traffic prediction systems.
- **Daily Life:** Voice assistants (like Siri, Alexa), recommendation systems (Netflix, YouTube), and smart home devices.
- **Education:** Personalized learning platforms and AI tutors are improving student engagement.

Q2. What's the difference between rational and irrational agents? Can irrational behavior ever be helpful?

Rational Agent

A **rational agent** is one that always tries to do the best thing it can—it makes decisions that maximize its chances of success based on the information it has. For example, a GPS navigation system choosing the fastest route is being rational.

Irrational Agent

On the other hand, an irrational agent doesn't always make the best decisions. Its actions might seem random or not really aimed at reaching a goal. At first, this might sound like a flaw, but surprisingly, irrational behavior can sometimes be useful.

Take exploration in learning, for example. A robot learning to walk might stumble around at first, trying weird moves that don't work. But some of those strange attempts might eventually lead to discovering better ways to walk. In other cases, like **video games or poker**, acting unpredictably can actually give you an edge because it keeps opponents guessing. And sometimes, AI systems use a bit of randomness to come up with **creative ideas**—like in music or art generation—where being too logical would make everything dull.

So, while we usually aim for rational behavior, a little irrationality now and then can help in unexpected ways.

Q.3. What Are the Biggest Challenges in AI Right Now? How Can We Fix Them?

AI is amazing, but it's not without problems. Here are five big challenges—and how we might tackle them:

1. AI Bias (When AI Gets Prejudiced)

- **Problem:** If AI learns from biased data, it can make unfair decisions (e.g., a hiring AI favoring men over women).
- **Fix:** Use more diverse data, audit AI systems for bias, and involve human oversight.

2. Privacy Risks (AI Knows Too Much)

- **Problem:** AI needs tons of data, but that can invade privacy (think facial recognition tracking people without consent).
- **Fix:** Stronger laws (like GDPR), anonymizing data, and using techniques like federated learning (where data stays on your device).

3. The "Black Box" Problem (AI Can't Explain Itself)

- **Problem:** Some AI (like deep learning) makes decisions in ways even experts don't fully understand.
- **Fix:** Develop "explainable AI" tools that break down decisions in simple terms.

4. Job Losses (Robots Taking Over Work)

- **Problem:** AI automation could replace many jobs (e.g., cashiers, drivers).
- **Fix:** Focus on reskilling workers, creating new AI-related jobs, and maybe even ideas like universal basic income.

5. Ethical Dilemmas (AI Making Moral Choices)

- **Problem:** Should a self-driving car prioritize passengers or pedestrians in a crash?
- **Fix:** Set clear ethical guidelines, involve philosophers and policymakers, and keep humans in the loop for critical decisions.

Q4. What is the Generate-and-Test method? What are its advantages and limitations?

Generate-and-Test is a simple problem-solving method used in Artificial Intelligence and computer science. The basic idea is just what it sounds like: you generate possible solutions to a problem, and then test each one to see if it works.

Here's how it works:

1. First, you generate a potential solution (this might be random or follow a rule).
2. Then, you test to see if it meets the conditions of a correct solution.
3. If it does, you're done. If not, you try again with a new one.

Example:

Let's say you're trying to guess a password that's a 4-digit number. You could use the generate-and-test method by trying "1234," then "5678," and so on—testing each one until the correct password is found.

Advantages:

- It's **very simple to understand and implement**.
- Useful when you **don't know much about the problem** in advance.
- Can work well in small or well-defined problem spaces.

Limitations:

- It can be **very slow** and **inefficient**, especially if the number of possibilities is huge.
- It doesn't learn or improve over time—it just keeps trying.
- It might **never find a solution** if the search space is too big or infinite.

Q5. How can a problem be represented as a state space using a graph or tree? Explain with an example.

In AI, many problems are represented as a state space, which is basically a collection of all the possible situations (or states) the problem can be in. We can show this using a graph or tree, where:

- Nodes represent different states.
- Edges (**arrows**) show the possible actions or moves that take you from one state to another.

This makes it easier to visualize and search for a solution.

Example: The 8-Puzzle Problem

The 8-puzzle is a game with 9 tiles (numbered 1 to 8, and one empty space). The goal is to arrange the tiles in a specific order by sliding them around.

- The initial state is the starting layout of the tiles.
- The goal state is the final, correct layout.
- Each move (sliding a tile into the empty space) changes the layout, creating a new state.
- The whole process can be drawn as a tree, starting from the initial state and branching out into all possible next states.

This tree or graph allows AI algorithms (like BFS, DFS, or A*) to search through the states and find the shortest or best path to the goal.

Why this is useful:

Once a problem is mapped this way, AI can systematically explore the space, evaluate paths, and choose the best strategy to reach the goal.

